

1. Artificial Intelligence (AI)

Definition

Artificial Intelligence (AI) is a field of computer science that focuses on creating computer systems that can perform tasks that normally require human intelligence.

Explanation

AI systems can understand information, recognize patterns, make decisions, solve problems, understand language, and interact with users. AI can be used in many areas such as healthcare, finance, education, transportation, and customer service.

AI is the broader field that includes technologies such as Machine Learning, Deep Learning, and Generative AI.

Example

A Google Maps system that predicts traffic and suggests a route is an example of AI.

2. Machine Learning (ML)

Definition

Machine Learning is a branch of AI in which computers learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every situation.

Explanation

Instead of writing rules for every possible situation, developers provide the machine-learning system with data. The system learns relationships and patterns from that data.

For example, a model can learn from thousands of emails that have been labeled as **spam** or **not spam**. After training, it can predict whether a new email is likely to be spam.

Example

A bank can use Machine Learning to detect unusual credit-card transactions and identify potentially fraudulent activity.

3. Deep Learning

Definition

Deep Learning is a subset of Machine Learning that uses multi-layered neural networks to learn complex patterns from large amounts of data.

Explanation

Deep-learning models contain multiple layers that transform input information into increasingly useful representations. They are particularly effective for tasks involving images, audio, video, and natural language.

Modern Generative AI and many Large Language Models are based on deep-learning techniques.

Example

A facial-recognition system can use a deep neural network to identify patterns in a person's face.

4. Generative AI

Definition

Generative AI is a type of AI that can create new content such as text, images, audio, video, or code based on instructions or input from a user.

Explanation

Traditional AI systems often classify, predict, or recommend something. Generative AI can **generate new output**.

Generative AI models learn patterns from large datasets and use those learned patterns to produce new content.

Example

ChatGPT can generate an explanation about Machine Learning when a user provides a prompt.

5. Relationship Between AI, ML, Deep Learning and Generative AI

Explanation

These technologies are related but are not exactly the same thing.

AI is the broadest field.

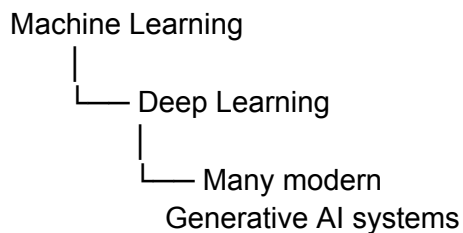
Machine Learning is a subset of AI that allows systems to learn from data.

Deep Learning is a subset of Machine Learning that uses deep neural networks.

Generative AI refers to AI systems designed to generate new content. Many modern Generative AI systems use deep-learning architectures such as Transformers.

Simple structure

Artificial Intelligence



Example

A system that predicts whether an image contains a cat is primarily a **classification** task.

A system that creates a new image of a cat from a text prompt is a **Generative AI** task.

6. Large Language Model (LLM)

Definition

A Large Language Model (LLM) is a deep-learning model trained on large amounts of text and other data to understand and generate human-like language.

Explanation

LLMs learn statistical patterns and relationships between tokens during training. When a user provides a prompt, the model processes the input and generates an output based on the patterns it learned.

LLMs can perform tasks such as:

- Question answering
- Summarization
- Translation
- Text generation
- Code generation
- Classification
- Information extraction
- Conversational interaction

Examples of modern LLM families include GPT, Gemini, and Claude.

Example

User:

Explain JavaScript to a beginner.

An LLM can generate a beginner-friendly explanation of JavaScript.

7. How Does an LLM Generate a Response?

Definition

LLM response generation is the process in which a model processes the user's input and predicts a sequence of tokens to produce an appropriate response.

Explanation

At a high level:

1. The user enters a prompt.
2. The text is converted into tokens.
3. The model processes those tokens together with the available context.
4. The model predicts likely next tokens.
5. The process continues token by token.
6. The generated tokens are converted back into readable text.

The model is not simply searching a database and copying an answer. It generates output based on patterns learned during training and the information available in the current context.

Example

Prompt:

What is React?

The model processes the prompt and generates tokens that form an explanation such as:

React is a JavaScript library used to build user interfaces...

8. Tokens

Definition

A token is a unit of text that an AI model processes. Depending on the tokenizer and language, a token can represent a word, part of a word, punctuation, or another piece of text.

Explanation

LLMs generally do not process an entire sentence exactly as humans see it. Text is first divided into tokens.

Token usage matters because AI APIs commonly measure input and output in tokens. Token limits also affect how much information can be processed in one request.

Tokens can therefore affect:

- API usage

- Cost
- Context capacity
- Amount of text that can be sent to a model

Example

The sentence:

I am learning AI.

is converted into a sequence of tokens by the model's tokenizer. The exact number and boundaries depend on the tokenizer being used.

9. Context Window

Definition

A context window is the amount of information, measured in tokens, that a model can consider as context during a particular interaction.

Explanation

The context can include things such as:

- User messages
- Previous conversation
- System instructions
- Documents
- Tool results
- Other information supplied to the model

A larger context window allows an application to provide more information to the model at once.

However, a context window is not the same thing as permanent memory. Information outside the available context may not be available to the model in that request.

Example

Suppose you build a document-analysis application. A user uploads a long document and asks:

Summarize this document.

The application needs to ensure that the relevant document content can fit within the model's available context, either directly or through techniques such as retrieval.

10. LLM Providers

Definition

An LLM provider is a company or organization that develops or provides access to AI models through products, platforms, or APIs.

Examples

- OpenAI
- Google
- Anthropic
- Meta
- Mistral AI
- Hugging Face

Different providers offer different models, capabilities, pricing structures, context limits, APIs, and deployment options.
